

Notes on Analysis

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Contents

I	Functional Analysis	1
1	Topological Vector Spaces	1
1.1	Some topology	1
1.2	Normed vector spaces	6

I. FUNCTIONAL ANALYSIS

1 Topological Vector Spaces

Real Analysis, Gerald Folland

1.1 Some topology

This section is concerned with some topological results which even a reader with some topological background may not be familiar with.

We begin by discussing the concept of a topological net. For nice enough spaces, we can learn a great deal about the topology of the space by studying sequences. Specifically this holds in first-countable spaces: spaces such that every point has a countable local basis. But in general this is not sufficient and leads to the generalization of a sequence to a *net*.

1.1.1 Definition

A **directed set** is a preorder (transitive and reflexive, not necessarily anti-symmetric) such that for every two elements α, β , there is a γ bounding them: $\alpha, \beta \leq \gamma$. A **net** on a topological space X is a (set-theoretic) map from a directed set I to X .

For example, $\mathbb{N}$ is a directed set and usual sequences from $\mathbb{N}$ are thus nets. Another important example is the set of neighborhoods of a point or subset of a topological space. That is, for a topological space X and a subset $S \subseteq X$, the set of all open subsets of X which contain S form a directed set, ordered by *inverse* inclusion. Inverse inclusion here is important, because usually we want $i \leq j$ when j in a sense “refines” i . We denote this directed set by $\mathcal{N}_S$, or $\mathcal{N}_x$ when S is the singleton $\{x\}$.

The product of directed sets is a directed set: if I, J are directed, then define a preorder on $I \times J$ by $(i, j) \leq (i', j')$ iff $i \leq i'$ and $j \leq j'$.

Note that a directed set is equivalent to a preorder which is filtered as a category. Then the product of directed sets just coincides with the product category of the two directed sets.

1.1.2 Definition

Let $(x_i)_{i \in I}$ be a net $I \rightarrow X$. Then for $S \subseteq X$, we say it is **eventually in S** if there is a $j_0 \in I$ such that $x_i \in S$ for all $i \geq j_0$. And it is **frequently in S** if for all $j \in I$ there is an $i \geq j$ such that $x_i \in S$. Clearly “eventually in S ” implies “frequently in S ”.

We say x_i **converges** to x , written $x_i \rightarrow x$, if x_i is eventually in every neighborhood of x . x is called a **limit** of x_i . It is a **cluster point** of x_i if x_i is frequently in every neighborhood of x .

1.1.3 Proposition

$x \in X$ is a limit point of $S \subseteq X$ iff there is a net in $S - x$ that converges to x . And x is in S 's closure iff there is a net in S that converges to x .

Proof

If x is a limit point of S , then let $\mathcal{N}$ be the directed set of neighborhoods of x , ordered by reverse inclusion. Then for $U \in \mathcal{N}$, choose $x_U \in U \cap (S - x)$ to form a net. Then $x_\bullet$ converges to x : if $\mathcal{U}$ is a neighborhood of x then for every $\mathcal{V} \in \mathcal{N}$ which is a subset of $\mathcal{U}$ we have

$$x_{\mathcal{V}} \in \mathcal{V} \cap (S - x) \subseteq \mathcal{U},$$

so that $x_\bullet$ is eventually in $\mathcal{U}$, and thus $x_\bullet \rightarrow x$.

Conversely if $x_\bullet$ is a net in $S - x$ converging to x , then for every neighborhood U of x there must be some i such that $x_i \in U$ and thus x is a limit point of S by definition.

Similar for the closure. ■

1.1.4 Proposition

A map between topological spaces $f: X \rightarrow Y$ is continuous at $x \in X$ iff for every net $x_\bullet$ converging to x , $f(x_\bullet)$ converges to $f(x)$.

Proof

If f is continuous at x , then let V be a neighborhood of $f(x)$, so $f^{-1}V$ contains a neighborhood of x : $x \in U \subseteq f^{-1}V$. So if $x_\bullet \rightarrow x$, then $x_\bullet$ is eventually in U and thus $f(x_\bullet)$ is eventually in V , so $f(x_\bullet) \rightarrow f(x)$ as desired.

Now if f is not continuous at x , there is a neighborhood $f(x) \in V$ such that $f^{-1}V$ is not a neighborhood of x (neighborhood meaning it contains an open neighborhood). That is, $x \notin f^{-1}V^\circ$, so equivalently $x \in \overline{f^{-1}V^c}$. By the previous proposition, there is a net $x_\bullet$ in $f^{-1}V^c$ that converges to x . Then $f(x_\bullet)$ is in V^c and thus not eventually in $f(x) \in V$, so it doesn't converge to $f(x)$. ■

Another nice property of nets is that convergence of nets categorize Hausdorff spaces.

1.1.5 Proposition

A topological space X is Hausdorff iff all nets over X converge to at most a single point.

Proof

If X is Hausdorff, then for $x \neq y$ there are disjoint open neighborhoods $x \in U, y \in V$. Then a net cannot be eventually in both U and V , by virtue of being directed (it would require an element in both U and V).

And conversely suppose X is not Hausdorff so there are $x \neq y$ without disjoint neighborhoods. Let $\mathcal{N}_x, \mathcal{N}_y$ be the directed sets of open neighborhoods of x, y respectively, and consider their product. We define a net on the product by choosing $x_{U,V} \in U \cap V$ for $U \in \mathcal{N}_x, V \in \mathcal{N}_y$. Then $x_{U,V}$ is easily seen to converge to both x and y . ■

We now discuss the concept of subnets, which is not as obvious a definition as it may seem.

1.1.6 Definition

A subset J of a preorder I is **cofinal** if for every $x \in I$ there is a $y \in J$ such that $x \leq y$.

Let $(x_i)_{i \in I}$ and $(y_j)_{j \in J}$ be nets to X over potentially different directed sets. Then $x_\bullet$ is a **subnet** of $y_\bullet$ if there is a monotone (order-preserving) map $f: I \rightarrow J$ whose image is cofinal in J , such that $x_i = y_{f(i)}$ for all $i \in I$.

Note that there is no requirement on f to be injective, so I could potentially be larger than J . Thus subsequences are subnets, but a subnet of a sequence need not be a subsequence!

A nice property, whose proof we omit due to disinterest (and not complexity!), is the following.

1.1.7 Proposition

A topological space X is compact iff every net has a convergent subnet.

This echoes the simpler result for metric spaces: a metric space is compact iff every sequence has a convergent subsequence (compactness is equivalent to sequential compactness). This is of course not true for general topological spaces, but it is when you swap sequences with nets.

1.1.8 Lemma

If $x_\bullet$ is a net $I \rightarrow X$, and $J \subseteq I$ is a cofinal subset, then the restriction of $x_\bullet$ to J is a subnet with precisely the same limits.

Proof

Pretty clear. ■

1.1.9 Proposition

$x \in X$ is a cluster point of a net $x_\bullet$ iff $x_\bullet$ has a subnet converging to x .

Proof

If $x \in X$ is a cluster point of $x_\bullet$, then consider the directed set $\mathcal{N}_x \times I$. Then for $(U, i) \in \mathcal{N}_x \times I$ there is an $x_j \in U$ for $j \geq i$; define $y_{(U,i)} = x_j$ and $f(U, i) = j$. Note that f is cofinal and $y_{U,i}$ converges to x . f is not necessarily monotone, but we can remove elements from our directed set to obtain the desired subnet (take a maximal subset for which f is monotone, and is easily seen to be cofinal).

The converse is easily seen. ■

1.1.10 Corollary

A topological space X is compact iff every net has a cluster point.

1.1.11 Definition

Let $\mathcal{F}$ be a collection of (set-theoretic) functions $X \rightarrow Y_i$, where Y_i are all topological spaces. Then the **initial** (or **weak**) topology on X is the coarsest topology (minimal with respect to inclusion) making each map $f \in \mathcal{F}$ continuous.

Dually if $\mathcal{F}$ is a collection of functions $X_i \rightarrow Y$ where each X_i is a topological space, the **final** topology on Y is the finest (maximal) topology making each $f \in \mathcal{F}$ continuous.

For $f \in \mathcal{F}$ to be continuous, then $f^{-1}U$ must be open for every open $U \subseteq Y_i$. Then the initial topology on X is just the topology generated by $f^{-1}U$ as f ranges over $\mathcal{F}$ and U over open subsets of Y_i . And the final topology on Y is given by requiring that $U \subseteq Y$ be open iff $f^{-1}U$ is open in X_i for each $f \in \mathcal{F}$.

For example, the topology on $\prod_i X_i$ is the initial topology on $\prod_i X_i$ given by the set of projections $\prod_i X_i \rightarrow X_i$. And the topology on $\prod_i X_i$ is the final topology given by the set of inclusions $X_i \rightarrow \prod_i X_i$. The quotient topology of a surjection $f: X \rightarrow Y$ is the final topology on the collection $\{f\}$.

1.1.12 Proposition

If X has the initial topology induced by $\mathcal{F}$, then a net $x_\bullet$ converges to $x \in X$ iff $f(x_\bullet)$ converges to $f(x)$ for all $f \in \mathcal{F}$.

Proof

Since each $f \in \mathcal{F}$ is continuous, the “only if” part is clear. Now if $f(x_\bullet) \rightarrow f(x)$ for all $f \in \mathcal{F}$, then let U be an open neighborhood of x . Since finite intersections of sets of the form $f_i^{-1}U_i$ for $f_i \in \mathcal{F}, U_i \subseteq X_i$ form a basis for X , we can assume $U = \bigcap_{i=1}^n f_i^{-1}U_i$, where U_i is an open neighborhood of $f(x)$ in X_i . Then there is some $\alpha_0 \in I$ such that for all $\alpha \geq \alpha_0$, $f_i(x_\alpha) \in U_i$ for all $i = 1, \dots, n$ (since there are finitely many i , we can bound the α_0 s chosen for $f_i(x_\bullet) \rightarrow f_i(x)$). Then $x_\alpha \in U$, so $x_\bullet$ is eventually in U as desired. ■

Nets provide a clean proof of Tychonoff’s theorem. We introduce some notation to ease the proof. If $\{X_\alpha\}_{\alpha \in A}$ is a collection of spaces, and $B \subseteq A$ a subset, we define

$$\pi_B: \prod_{\alpha \in A} X_\alpha \rightarrow \prod_{\beta \in B} X_\beta,$$

by mapping $x \in \prod_{\alpha \in A} X_\alpha$ to the restriction of x to B (viewing the product as a certain set of maps $A \rightarrow \prod_{\alpha \in A} X_\alpha$). In particular, when B is a singleton π_B is essentially the same as projection onto X_β . For $p \in \prod_{\beta \in B} X_\beta$ and $q \in \prod_{\gamma \in C} X_\gamma$, we say $p \leq q$ if $B \subseteq C$ and p is the restriction of q to B ($\pi_B(q) = p$). We also say that q *extends* p (since it does as a function).

1.1.13 Theorem (Tychonoff)

The product of an arbitrary collection of compact spaces is compact.

Proof

Let $\{X_\alpha\}_{\alpha \in A}$ be compact, and let $X = \prod_{\alpha \in A} X_\alpha$. We want to say that X is compact. We do this by showing any net $(x_i)_{i \in I}$ has a cluster point.

Define

$$P = \bigcup_{B \subseteq A} \left\{ p \in \prod_{\beta \in B} X_\beta \mid p \text{ is a cluster point of } (\pi_B(x_i))_i \right\}.$$

This is non-empty since when B is a singleton, $(\pi_B(x_i))$ has a cluster point as each X_α is compact. And P is ordered by extension as above.

We now show that P satisfies the condition for Zorn’s lemma and thus has a maximal element. Indeed, let $\{p_\ell\}_{\ell \in L} \subseteq P$ be a chain. Suppose $p_\ell \in \prod_{\alpha \in B_\ell} X_\alpha$.

Define $B^* = \bigcup_{\ell \in L} B_\ell$, and define $p^* \in \prod_{\alpha \in B^*} X_\alpha$ by $p^*(\alpha) = p_\ell(\alpha)$ when $\alpha \in B_\ell$. This is well-defined since if $\alpha \in B_\ell \cap B_{\ell'}$ then $p_\ell(\alpha) = p_{\ell'}(\alpha)$ since by linearity we can assume $p_\ell \leq p_{\ell'}$, and thus $p_\ell = p_{\ell'}|_{B_\ell}$. Clearly $p_\ell \leq p^*$ for all $\ell \in L$, so it is an upper bound if we can show it is in P .

So we need to show that p^* is a cluster point for $\pi_{B^*}(x_\bullet)$. Let U be a neighborhood of p^* , we can assume it is a basis set, so $U = \prod_{\alpha \in B^*} U_\alpha$ where all but finitely many U_α are equal to X_α . Let $\alpha_1, \dots, \alpha_n$ enumerate the indices where $U_{\alpha_i} \neq X_{\alpha_i}$. Since $\{p_\ell\}$ and thus B_ℓ are linearly ordered, these α_i s are all contained in some B_ℓ . Then $\pi_{B_\ell}(x_\bullet)$ is frequently in $\pi_{B_\ell}(U)$ as it is a neighborhood of $p_\ell \in P$. Since all the other α s have $U_\alpha = X_\alpha$, clearly $\pi_{B^*}(x_\bullet)$ is in U when $\pi_{B_\ell}(x_\bullet)$ is in $\pi_{B_\ell}(U)$, so $\pi_{B^*}(x_\bullet)$ is frequently in U . Thus p^* is a cluster point of $\pi_{B^*}(x_\bullet)$, so $p^* \in P$ and thus is an upper bound for $\{p_\ell\}$.

Thus by Zorn, P has a maximal element $p \in \prod_{\beta \in B} X_\beta$. We claim that $B = A$, so that $x_\bullet$ has a cluster point and we will finish. If not, then let $\gamma \in A - B$, and let $\pi_B(x_{i(j)})_{j \in J}$ be a subnet of $\pi_B(x_i)_{i \in I}$ converging to p . By compactness, there is a subnet $\pi_\gamma(x_{i(j(k))})_{k \in K}$ of $\pi_\gamma(x_{i(j)})_{j \in J}$ converging to some $p_\gamma \in X_\gamma$.

Then there is a unique element $q \in \prod_{\alpha \in B \cup \gamma} X_\alpha$ extending p_γ, p . The net $\pi_{B \cup \gamma}(x_{i(j(k))})_{k \in K}$ converges to q , and thus q is a cluster point of $\pi_{B \cup \gamma}(x_i)$, which contradicts p ’s maximality. ■

1.1.14 Definition

If X is a topological space, let $C(X)$ denote the set of continuous complex-valued maps $X \rightarrow \mathbb{C}$ (sometimes

this will be used for real-valued functions, too). Note that $C(X)$ is a complex algebra under pointwise addition and multiplication. If X is compact, we define a norm on $C(X)$, the **uniform norm** by

$$\|f\|_u = \sup_{x \in X} |f(x)|.$$

This is well-defined (not infinite) because $|f(x)|$ is a continuous map from a compact space to the reals and thus is bound.

The uniform norm is easily seen to be norm, and thus turns $C(X)$ into a topological complex algebra. Furthermore, $C(X)$ is complete.

1.1.15 Definition

A subset $\mathcal{F} \subseteq C(X)$ is **equicontinuous at** $x \in X$ if for every $\varepsilon > 0$ there is a neighborhood U of x such that $|f(x) - f(y)| < \varepsilon$ for all $f \in \mathcal{F}, y \in U$. It is **equicontinuous** if it is equicontinuous at every point in X .

We also say that $\mathcal{F}$ is **pointwise bounded** if $\{f(x) \mid x \in X, f \in \mathcal{F}\}$ is a bounded complex set.

We state the following theorems without proof (again out of disinterest),

1.1.16 Theorem (Arzela-Ascoli, I)

Let X be a compact Hausdorff space, and $\mathcal{F} \subseteq C(X)$ an equicontinuous pointwise-bounded set. Then $\mathcal{F}$ is totally bounded in the uniform metric, and $\mathcal{F}$'s closure in $C(X)$ is compact.

1.1.17 Definition

A topological space is said to be **σ -compact** if it is the union of countably many compact subsets. It is **locally compact** if every point has a compact neighborhood (is contained in an open neighborhood contained in a compact set).

1.1.18 Theorem (Arzela-Ascoli, II)

Let X be a σ -compact locally compact Hausdorff space. If $\{f_n\}$ is equicontinuous and pointwise bounded in $C(X)$, there there is an $f \in C(X)$ and a subsequence of f_n which converges uniformly to f on compact sets.

We also state the following without proof (see e.g. Folland for a complete proof, this one is a little tedious).

1.1.19 Definition

A subset $\mathcal{A} \subseteq C(X, Y)$ **separates points** if for all $x \neq y$ in X , there is an $f \in \mathcal{A}$ such that $f(x) \neq f(y)$.

1.1.20 Theorem (Stone-Weierstrass)

Let X be compact Hausdorff. If $\mathcal{A}$ is a closed subalgebra of $C(X, \mathbb{R})$ which separates points, then either

$\mathcal{A} = C(X, \mathbb{R})$ or $\mathcal{A} = \{f \in C(X, \mathbb{R}) \mid f(x_0) = 0\}$ for some $x_0 \in X$. The first case holds iff $\mathcal{A}$ contains the constant functions.

1.1.21 Corollary (Weierstrass approximation theorem)

Let X be a compact subset of $\mathbb{R}^n$, then the set of polynomials on $\mathbb{R}^n$ (meaning the polynomial algebra $\mathbb{R}[x_1, \dots, x_n]$) is dense in $C(X, \mathbb{R})$.

Proof

Let $\mathcal{A} = \mathbb{R}[\bar{x}]$. Then $\mathcal{A}$ is a subalgebra of $C(X, \mathbb{R})$ which separates points and contains the constant functions. Its closure is also a subalgebra (by continuity of multiplication and addition), and thus by Stone-Weierstrass is $C(X, \mathbb{R})$, as desired. ■

For the complex case we must also have that $\mathcal{A}$ is closed under complex conjugation.

1.1.22 Theorem (Stone-Weierstrass, complex case)

Let X be compact Hausdorff. If $\mathcal{A}$ is a closed subalgebra of $C(X)$ which separates points and is closed under complex conjugation, then either $\mathcal{A} = C(X)$ or $\mathcal{A} = \{f \in C(X) \mid f(x_0) = 0\}$ for some $x_0 \in X$.

Proof

Since $\operatorname{Re} f = (f + \bar{f})/2$ and $\operatorname{Im} f = (f - \bar{f})/2i$, the set $\mathcal{A}_{\mathbb{R}}$ of real and imaginary parts of functions from $\mathcal{A}$ satisfies the conditions of the real Stone-Weierstrass theorem, thus $\mathcal{A}_{\mathbb{R}}$ is either $C(X, \mathbb{R})$ or $\{f \in C(X, \mathbb{R}) \mid f(x_0) = 0\}$. Since $\mathcal{A} = \{f + ig \mid f, g \in \mathcal{A}_{\mathbb{R}}\}$ the result follows. ■

1.2 Normed vector spaces

In this section K denotes a field, either $\mathbb{R}$ or $\mathbb{C}$.

1.2.1 Definition

Let X be a vector space over K . A **seminorm** on X is a function $\|\bullet\| : X \rightarrow \mathbb{R}_{\geq 0}$ that is

- (1) **subadditive:** $\|x + y\| \leq \|x\| + \|y\|$ (also called the **triangle inequality**),
- (2) $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in K$.

A **norm** is a seminorm which is zero only at $0 \in X$.

Norms define metrics via $(x, y) \mapsto \|x - y\|$ and thus topologies. A normed vector space which is complete with respect to this metric is called a **Banach space**.

A series $\sum_n x_n$ with $x_n \in X$ is said to be **absolutely convergent** if $\sum_n \|x_n\|$ converges.

1.2.2 Theorem

X is complete iff every absolutely convergent series converges.

Proof

If $\sum_n x_n$ is absolutely convergent then its partial sums are Cauchy:

$$\left\| \sum_{n=1}^N x_n - \sum_{n=1}^M x_n \right\| = \left\| \sum_{n=M+1}^N x_n \right\| \leq \sum_{n=M+1}^N \|x_n\| \rightarrow 0 \text{ as } M, N \rightarrow \infty.$$

Thus $\sum x_n$ is convergent.

Conversely if absolutely convergent series converge, then let $\{x_n\}$ be Cauchy. Then let $n_1 < n_2 < \dots$ be such that $\|x_n - x_m\| < 2^{-j}$ when $n, m \geq n_j$. Define $y_j = x_{n_j} - x_{n_{j-1}}$ and $y_1 = x_{n_1}$ so that $\sum_1^k y_j = x_{n_k}$, and

$$\sum \|y_j\| \leq \|y_1\| + \sum 2^{j-1} < \infty$$

so that $\sum y_j$ is absolutely convergent and thus convergent. Since $\sum_1^k y_j = x_{n_k}$, x_{n_k} is convergent, and thus x_n is as well (a Cauchy sequence with a convergent subsequence is convergent). ■

Examples of Banach spaces include L^p spaces over arbitrary measure spaces.

The product of normed vector spaces is normed: $\|(x, y)\| = \max\{\|x\|, \|y\|\}$. And the quotient of a normed vector space has a natural seminorm: define the quotient norm on X/Y by $\|x + Y\| = \inf_{y \in Y} \|x + y\|$. This is a norm precisely when $Y \subseteq X$ is closed (since $\inf_{y \in Y} \|x + y\| = 0$ means that there is a sequence of vectors from Y converging to x).

A linear map between normed spaces $T: X \rightarrow Y$ is **bounded** if there is a $C \geq 0$ such that

$$\|Tx\| \leq C \|x\|, \quad \text{for all } x \in X.$$

1.2.3 Proposition

If $T: X \rightarrow Y$ is a map between normed vector spaces, the following are equivalent:

- (1) T is Lipschitz-continuous,
- (2) T is continuous,
- (3) T is continuous at 0,
- (4) T is bounded.

Proof

If T is continuous at 0, then it must map an open ball around 0, say $B_\varepsilon(0)$, to the open ball $B_1(0)$ (the former in X , the latter in Y). Then $\|Tx\| \leq 1$ when $\|x\| < \varepsilon$. Then if $\|x\| < a\varepsilon$ we have that $\|Tx\| = a \|T(x/a)\| < a\varepsilon$. Thus

$$\|Tx\| \leq \varepsilon^{-1} \|x\|,$$

so T is bounded.

And if T is bounded, say by C , then $\|Tx_1 - Tx_2\| \leq C \|x_1 - x_2\|$ so it is Lipschitz-continuous. ■

Define $L(X, Y)$ to be the vector space of all bounded linear maps $X \rightarrow Y$. We define a norm on it, the **operator norm** by

$$\|T\| = \sup \{\|Tx\| \mid \|x\| = 1\} = \sup \left\{ \frac{\|Tx\|}{\|x\|} \mid x \neq 0 \right\} = \inf \{C \mid \|Tx\| \leq C \|x\| \text{ for all } x\}.$$

Notice that between finite-dimensional spaces all linear maps are bounded.

1.2.4 Proposition

If Y is complete, so is $L(X, Y)$.

Proof

Let $\{T_n\}$ be Cauchy in $L(X, Y)$, then for $x \in X$ we have

$$\|T_n x - T_m x\| \leq \|T_n - T_m\| \|x\|,$$

so that $T_n x$ is Cauchy in Y and thus converges to some $Tx \in Y$. T is then linear (by linearity of limits), $\|T_n - T\| \rightarrow 0$, and $\|T\| = \lim \|T_n\| < \infty$ so $T \in L(X, Y)$. ■

Notice that if $T \in L(X, Y)$, $S \in L(Y, Z)$ then

$$\|STx\| \leq \|S\| \|Tx\| \leq \|S\| \|T\| \|x\|,$$

so that $\|ST\| \leq \|S\| \|T\|$. Thus we can form a category of normed vector spaces with bounded linear maps between them as morphisms. In particular, $L(X, X)$ is an algebra.

An isomorphism in this category is a bijective linear map $T: X \rightarrow Y$ such that both T and T^{-1} are bounded. This means that $\|Tx\| \leq C \|x\|$ for some $C > 0$ and also $\|Tx\| \geq D \|x\|$ for some $D > 0$.

1.3 Linear functionals

For a complex number $z \in \mathbb{C}$ we define $\operatorname{sgn} z = z/|z|$, so that $|z| = \overline{\operatorname{sgn} z} z$.

1.3.1 Definition

Let X be a normed vector space over K . Define its **dual space** by $X^* = L(X, K)$, the space of bounded linear functionals.

1.3.2 Proposition

Let X be a complex vector space, and f a complex linear functional on X . Let $u = \operatorname{Re} f$, then u is a real linear functional on X and $f(x) = u(x) - iu(ix)$. Conversely if u is a real linear functional on X , then $f(x) = u(x) - iu(ix)$ is a complex linear functional. Furthermore $\|u\| = \|f\|$.

Proof

Since $\operatorname{Im} f(x) = -\operatorname{Re}(if(x)) = -\operatorname{Re}(f(ix)) = -u(ix)$ we see that $f(x) = u(x) - iu(ix)$ as desired. The rest is easily verified. And since $|u(x)| = |\operatorname{Re} f(x)| \leq |f(x)|$ we see that $\|u\| \leq \|f\|$. Let $\alpha = \operatorname{sgn} f(x)$, so $|f(x)| = \alpha f(x) = f(\alpha x) = u(\alpha x)$ (since $f(\alpha x)$ is real). Thus

$$|f(x)| \leq \|u\| \|\alpha x\| = \|u\| \|x\| \implies \|f\| \leq \|u\|. \quad \blacksquare$$